Phosphoprotein regulation

Which proteins carry the regulated phosphosites, and whether their sites move together or against each other.

AT A GLANCE

Where	Phospho → Phosprot Regulation
Data level	Phosphosite throughout. The page aggregates site-level results per protein; it does not read a protein matrix and performs no protein normalisation.
Needs	An annotated phospho dataset with Protein_group and PTM_Collapse_key, and two different conditions with at least two replicates each.
Direction	$\log_2$ FC = **Condition 1** − **Condition 2**. "Up" means higher in **Condition 1**.
Both axes	Non-negative. The figure occupies one quadrant, not four.
Figure	Interactive, drawn in a frame on the page. There is no image download; **Open Plot** opens it on its own.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 The question this page exists for

A volcano plot of phosphosites answers which sites changed; it does not answer which *proteins* the changes landed on, and that is usually what a signalling experiment is asking. This page collapses the site-level result onto proteins and keeps the two directions apart instead

of letting them cancel: for every protein it adds up the fold changes of its significant up sites, adds up the magnitudes of its significant down sites, and plots one against the other. A protein whose sites all move the same way lands on one of the axes; a protein whose sites disagree lands out in the field, and that disagreement is a result rather than noise.

> **NOTE**
>
> This is not protein-abundance correction. A site whose protein was itself induced appears here exactly like a site whose occupancy rose at constant protein level, because no protein matrix is consulted. To investigate the two, compare site-level changes with independently measured protein abundance. Collapsing sites alone does not provide that protein measurement and does not estimate absolute site occupancy.

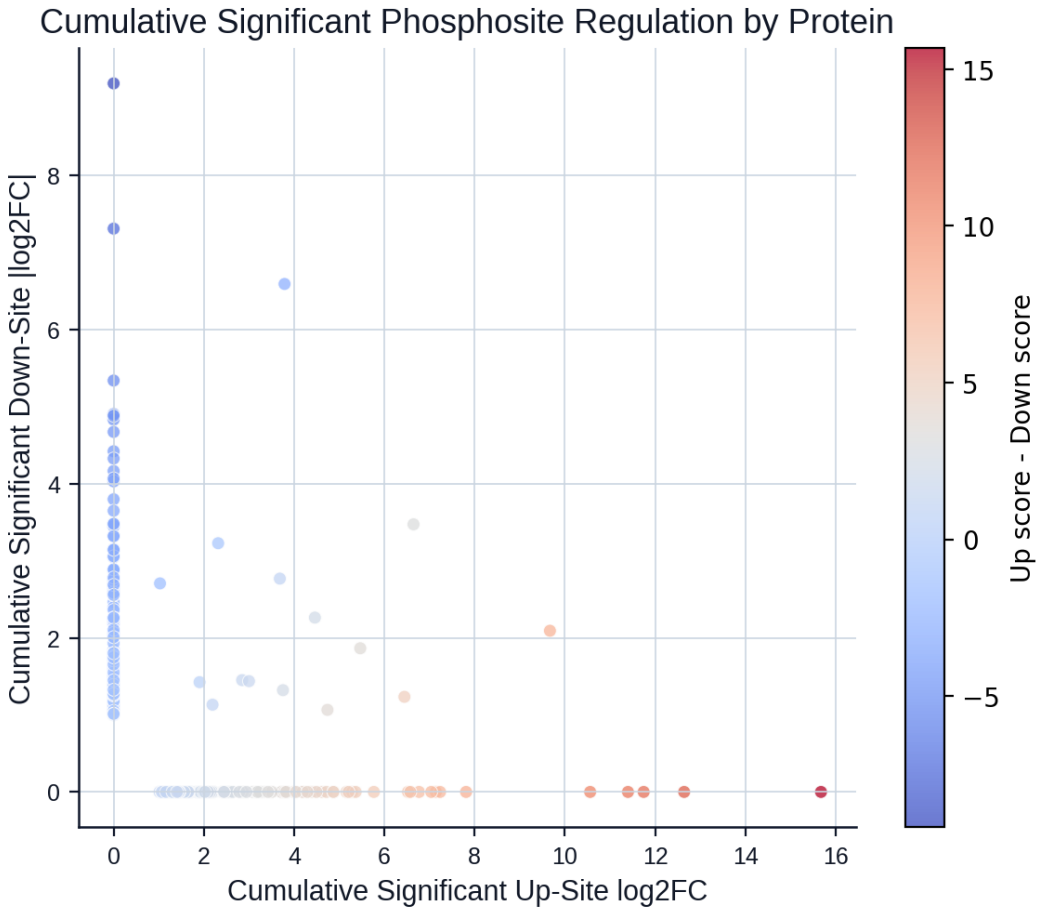


Figure 1.1. WT against GRK on the example dataset, at the default thresholds. Each point is one protein with at least one significant site: x is the sum of the $\log_2$ fold changes of its significant up sites, y the sum of the magnitudes of its significant down sites, and colour is $x - y$ on the blue-to-red scale at the right. Most points sit on one of the two axes — a dense row along $y = 0$ and a column along $x = 0$ — and only a scattering lies away from both. That describes this example; mixed directions at different sites can also be biologically meaningful.

Reading it

- One point, one protein. Only proteins with at least one significant site are drawn; the rest are absent from the figure and from the table.
- The x axis, Cumulative Significant Up-Site log2FC: the sum of $\log_2$ FC over the sites of that protein classified Upregulated. Zero when it has none, never negative.
- The y axis, Cumulative Significant Down-Site |log2FC|: the sum of the *magnitudes* of $\log_2$ FC over its Downregulated sites. Zero when it has none, never negative.
- Where a point sits. On the x axis: every significant site moved up. On the y axis: every one moved down. Away from both: significant sites in both directions, the distance from the origin being the total regulated signal on that protein, and the diagonal $x = y$ the balance point.
- Colour, from the scale Up score - Down score $= x - y$: red for net upward, blue for net downward, grey for balanced. It adds nothing the coordinates do not, but it makes the diagonal readable.
- Hover: the protein group, the number of significant sites in each direction, and those sites with their fold changes, up to **Max Sites in Table**.

> **CHANGED IN 1.0.0**
>
> Earlier documentation of this page stated that the function computes $\log_2$ fold changes and p-values for all phosphosites but applies no significance cutoff. That is no longer the case. **P-value Threshold** and **log2 FC Threshold** now decide which sites enter the sums, and a protein with no significant site is not drawn at all. A figure made with the old behaviour is not comparable to one made now.

2 Controls

Controls on this page

Control	What it does	Default	Where
Condition 1	The first condition of the comparison, and the minuend: $\log_2$ FC = **Condition 1** − **Condition 2**.	first condition in the annotation	page
Condition 2	The second condition. Choosing the same condition twice draws nothing and says so.	first condition in the annotation	page

Controls on this page (continued)

Control	What it does	Default	Where
P-value Threshold	A site is significant when its p-value is strictly below this. The value is the adjusted p-value unless **Use Uncorrected P-values** is on. Clamped to $[10^{-12}, 1]$.	0.05	page
log2 FC Threshold	A site must also exceed this in magnitude: above $+t$ to count as up, below $-t$ to count as down. 0 keeps the p-value criterion alone.	1	page
Test Type	Unpaired (Student's t, equal variance), Unpaired (Welch, unequal variance) or Paired t-test (matched samples); the last opens the pairing editor and needs every sample paired before a figure is drawn.	Unpaired (Student's t)	page
Use Uncorrected P-values	Compares raw p-values against **P-value Threshold** instead of Benjamini–Hochberg adjusted ones. This admits more sites, not fewer.	off	page
Show Phosphosites in Table	Whether the site identifiers and their fold changes are listed. Off leaves the four site columns empty and keeps the counts.	on	page
Max Sites in Table	How many site identifiers to list per protein and direction before the rest are summarised as … (+n more).	20	page
Width (cm), Height (cm), DPI	Present on the page, but the interactive figure this tab renders ignores them: it is sized to the frame it is drawn in.	20 × 12, 220	page

In the plot appearance panel

Control	What it does	Default	Where
Points	Size, opacity and outline of the markers. The fill is not offered: the points are coloured from $x - y$ through the colour scale.	size 8, opacity 0.7, outline 0.5	appearance
Title, axis labels	Text, size, weight, colour and position, and whether the title is drawn. The plot writes Cumulative Significant Phosphosite Regulation by Protein over the two axis titles quoted above.	shown	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Axis frame, grid, background	Which axes are drawn, the grid over them, paper colour and the preset themes.	default	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

STEP ONE: THE SITE-LEVEL TEST

Every phosphosite is tested between the two conditions on the annotated log2 matrix, by exactly the procedure the volcano plot uses and described there: a row needs two finite values in each condition, the fold change is $\overline{x}_{\text{cond 1}} - \overline{x}_{\text{cond 2}}$, the p-value comes from the two-sided t-test **Test Type** selects, and Benjamini–Hochberg adjusts it unless **Use Uncorrected P-values** is on. A site is Upregulated when $p_{\text{adj}} <$ **P-value Threshold** and $\log_2 \text{FC} >$ **log2 FC Threshold**, Downregulated when $p_{\text{adj}} <$ **P-value Threshold** and $\log_2 \text{FC} <$ **−log2 FC Threshold**, and Not significant otherwise.

STEP TWO: THE COLLAPSE ONTO PROTEINS

The labelled sites are grouped by Protein_group. A group with no Upregulated and no Downregulated site is discarded. For the rest, with U the set of up sites and D the set of down sites of that protein,

$$x = \sum_{i \in U} \log_2 \text{FC}_i, \qquad y = -\sum_{i \in D} \log_2 \text{FC}_i, \qquad \text{colour} = x - y.$$

Both sums are over significant sites only, so a protein with twenty unchanged sites and one large change is placed by that one change. Because the members of U are all above $+t$ and those of D all below $-t$, both coordinates are non-negative, and a protein with sites in only one direction sits exactly on an axis.

Edge cases

- If no site anywhere passes both thresholds, the figure is drawn empty and carries the sentence No phosphosites pass the selected significance and fold-change thresholds. This is the answer to a legitimate question and not an error; earlier releases failed here instead, and the removed explanation box on the page is replaced by this section.
- Selecting the same condition in both boxes draws nothing and says so; so does a paired test whose samples are not fully paired.

- A site the test could not evaluate — fewer than two values in a condition — is labelled Unavailable and enters neither sum. Duplicate site identifiers are kept apart during the test, so a protein carrying the same PTM_Collapse_key twice contributes both rows.
- The page reads the annotated log2 matrix, not the filtered or imputed one.

> **EASY TO GET WRONG**
>
> Both axes are *sums*, so they grow with the number of significant sites as well as with the size of the changes. A protein far out along x may carry one site that moved eight-fold or eight sites that each moved a little, and the figure does not distinguish them: the hover and the Upregulated_count column do. Nor is either coordinate comparable to a fold change — adding $\log_2$ ratios multiplies the ratios, and the result is a score for ranking proteins within one figure, not a quantity to report.

4 The table below the plot

Columns in this table

Column	What it holds
Protein_group	The protein the point stands for.
Upregulated_count	Its significant up sites.
Downregulated_count	Its significant down sites.
Upregulated_log2FC_sum	The x coordinate.
Downregulated_log2FC_sum	The y coordinate, already made positive.
Upregulated_sites, Downregulated_sites	The site identifiers in each direction, separated by semicolons and truncated at **Max Sites in Table**. Empty when **Show Phosphosites in Table** is off.
Upregulated_site_effects, Downregulated_site_effects	The same identifiers with each site's $\log_2$ FC in brackets, signed.
x, y, color_val	The plotted coordinates and $x - y$, repeated as their own columns so the figure and the table cannot drift apart.

Rows are ordered by protein group and download as CSV. The figure has no image download on this tab: **Open Plot** opens the interactive version on its own, where it can be zoomed and saved from the plot's own toolbar.

A protein can genuinely carry sites changing in opposite directions. Do not adjust thresholds merely to move points onto the axes or increase their number. These coordinates are sums: a protein with many quantified sites has more opportunities to accumulate a large score than

one with few. They are neither a protein fold change nor a measure of kinase activity. Inspect the contributing sites, their replicate values and the multiple-testing setting before drawing a biological conclusion.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = phosprot_regulation(data3, meta2, "WT", "GRK", in_pval=0.05, in_log2fc=1)
```

R

```
fig <- phosprot_regulation(data3, meta2, "WT", "GRK", in_pval = 0.05, in_log2fc = 1)
```

The arguments are the page's controls under other names: in_pval and in_log2fc are the two thresholds, uncorrected is **Use Uncorrected P-values**, paired and test_type carry **Test Type**, and max_hover_sites and show_phosphosites govern the site lists in the hover and the table. What comes back is the interactive figure itself — the one the page hides behind **Open Plot** — and phosprot_regulation_data returns the same per-protein table without drawing anything.

6 Further reading

Wu et al., absolute phosphorylation stoichiometries (Nature Communications, 2015) discusses why phosphosite abundance alone does not distinguish occupancy from protein abundance. The plot here summarizes measured site changes; it does not perform that separation.

Behind this page

Function	What it produces
phosprot_regulation	The scatter, in the Python package
phosprot_regulation_data	The per-protein table the scatter plots
phosprot_regulation_table	The same table in the web application
phosprot_regulation_html	The interactive figure the page renders
phosprot_regulation_png	The static version of the same figure